

Special39 / competency practice — Electricity: Materials, Geometry and Design — Original Practice Paper 04

CLASS 10 • ELECTRICITY • ORIGINAL PRACTICE

Original Class 10 Electricity • 39 questions • 80 marks • competency practice

Original chapter practice — not an official examination paper, a prediction, or an official chapter weightage.

39 questions • 80 marks • 180 minutes

An original 39-question chapter practice paper using material comparisons, wire geometry and experimental inference alongside the full Class X Electricity chapter. Not an official paper, prediction or chapter-weightage claim.

Paper structure

Section A | Questions 1–20 | $20 \times 1 = 20$ marks

Section B | Questions 21–26 | $6 \times 2 = 12$ marks

Section C | Questions 27–33 | $7 \times 3 = 21$ marks

Section D | Questions 34–36 | $3 \times 5 = 15$ marks

Section E | Questions 37–39 | $3 \times 4 = 12$ marks

Instructions

Attempt all questions. There are no internal OR choices. Sections A–E form the question paper. Detailed solutions and marking guidance begin only after Question 39; source-pattern and competency rationale follow in a separate appendix.

Instructions

All 39 questions are compulsory. There are no internal OR choices. Section A: Q1–20, 1 mark each; Section B: Q21–26, 2 marks each; Section C: Q27–33, 3 marks each; Section D: Q34–36, 5 marks each; Section E: Q37–39, 4 marks each.

For Q1–16 choose one correct option. For Q17–20 use the common assertion–reason options listed below. Show reasoning, formulae, substitutions and units in written answers; subpart marks are stated.

Unless specifically varied, use ideal fixed-voltage sources, negligible resistance of connecting wires and contacts, ideal ammeters (zero resistance), and ideal voltmeters (no current). Test conductors are uniform and ohmic at constant temperature. In heater calculations use the stated constant operating resistance/resistivity; disregard changes during warming. Energy losses outside a named resistive element are neglected unless stated.

Redrawing conserves conducting-material volume where stated; it does not change resistivity. Area means conducting cross-sectional area, not insulation area. Diagrams are schematic, not to scale; use their labels and axis values. Symbols A and R in assertion-reason items mean Assertion and Reason, not area and resistance.

- A: Both A and R are true, and R correctly explains A.
- B: Both A and R are true, but R does not correctly explain A.
- C: A is true, but R is false.
- D: A is false, but R is true.

Section A

Question 1 | 1 mark

special-39-04-q01

Which statement correctly describes electric current and the connection of an ammeter used to measure the current through a resistor? [1 mark]

- A. Current is energy transferred each second; connect the ammeter in series.
- B. Current is charge passing a cross-section each second; connect the ammeter in series.
- C. Current is charge passing a cross-section each second; connect the ammeter in parallel.
- D. Current is energy transferred per coulomb; connect the ammeter in parallel.

Question 2 | 1 mark

special-39-04-q02

Take V and Q as positive. In trial X, a source transfers energy W while moving charge Q through a potential difference V . In trial Y, the potential difference is $3V$ and the transferred charge is $Q/2$. What is the electrical work done in trial Y? [1 mark]

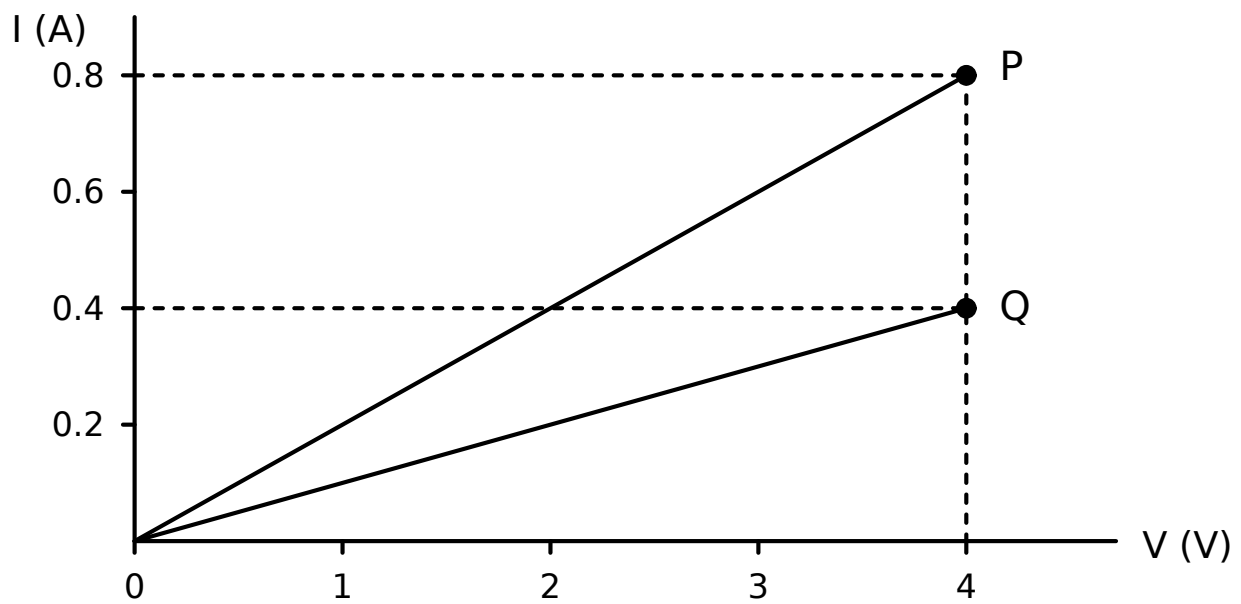
- A. $W/6$
- B. $6W$
- C. $3W/2$
- D. $2W/3$

Question 3 | 1 mark

special-39-04-q03

The figure gives measured I - V lines for wires P and Q at the same temperature. P has length L and conducting area A ; Q has length $4L$ and conducting area A . Which material-resistivity relation is supported after allowing for this unequal geometry? [1 mark]

Measured I-V lines at the same temperature



P: length L , area A ; Q: length $4L$, area A

- A. P has twice the resistivity of Q.
- B. Q has twice the resistivity of P.
- C. They have equal resistivity because their areas are equal.
- D. Q has four times the resistivity of P.

Question 4 | 1 mark

special-39-04-q04

A uniform sample X has length L and area A . Sample Y has length $2L$ and area $2A$. At the same temperature both have resistance $8\ \Omega$. Which conclusion follows? [1 mark]

- A. Y has half the resistivity of X.
- B. Y has twice the resistivity of X.
- C. The two samples have equal resistivity, but this alone does not prove identical composition.
- D. Equal resistance proves identical chemical composition.

Question 5 | 1 mark

special-39-04-q05

A cable maker doubles only the outer diameter of the plastic insulation around a metal core. Core material, core diameter and core length stay unchanged. Which statement about the resistance between the core ends is correct? [1 mark]

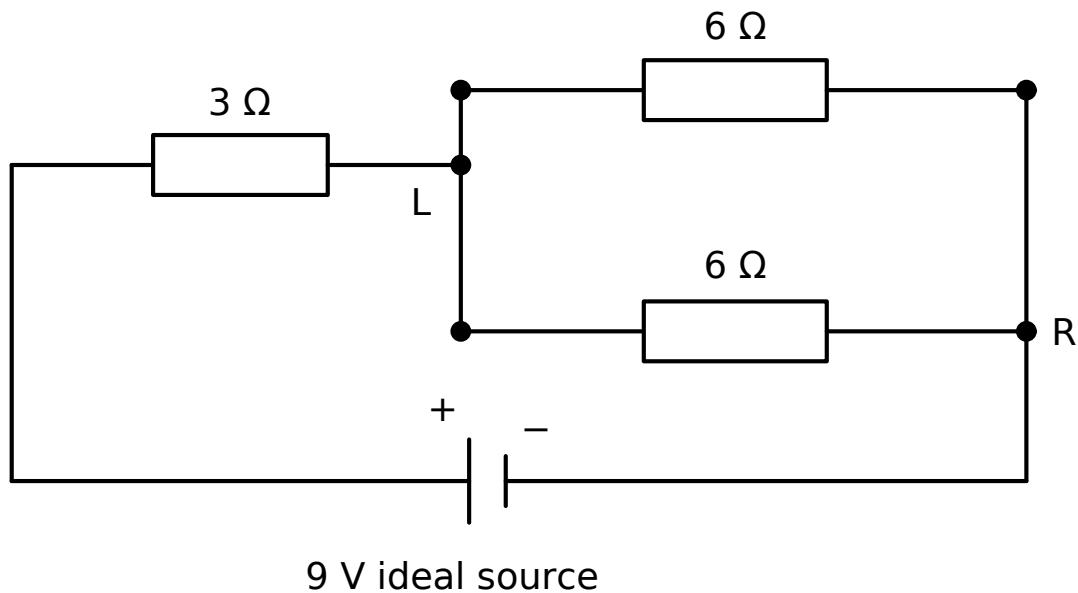
- A. It is unchanged because the conducting cross-section is unchanged.
- B. It becomes one-quarter because the outer diameter doubles.
- C. It becomes one-half because insulation is thicker.

D. It doubles because the cable is wider.

Question 6 | 1 mark

special-39-04-q06

The circuit in the figure is connected to an ideal 9 V source. Which pair gives the source current and the current through the upper $6\ \Omega$ resistor, respectively? [1 mark]

Circuit for Question 6

- A. 3 A, 1.5 A
- B. 1.5 A, 1.5 A
- C. 0.75 A, 0.75 A
- D. 1.5 A, 0.75 A

Question 7 | 1 mark

special-39-04-q07

In a steady series circuit, 18 C passes into a heater in 6 s. A student says that only 12 C will leave it in those 6 s because the rest becomes heat. Which correction is valid? [1 mark]

- A. 12 C leaves, but the lost charge becomes light rather than heat.
- B. 18 C leaves; electrical energy is transferred but charge is not consumed.
- C. No charge leaves a resistance.
- D. 36 C leaves because electrons multiply in a hot wire.

Question 8 | 1 mark

special-39-04-q08

A wire has conducting cross-sectional area $0.35\ \text{mm}^2$. Express this area in m^2 for use in $R = \rho L/A$. [1 mark]

- A. $3.5 \times 10^{-4} \text{ m}^2$
- B. $3.5 \times 10^{-6} \text{ m}^2$
- C. $3.5 \times 10^{-7} \text{ m}^2$
- D. $3.5 \times 10^{-8} \text{ m}^2$

Question 9 | 1 mark

special-39-04-q09

A class records (V, I) as (1.2 V, 0.20 A), (2.4 V, 0.40 A), (3.6 V, 0.60 A) and (4.8 V, 0.60 A). A repeat at 4.8 V gives 0.80 A. Which claim is best supported? [1 mark]

- A. The repeat restores the common 6Ω pattern; the first last-row reading should be investigated.
- B. Every reading proves resistance is 8Ω .
- C. Current can never exceed 0.60 A.
- D. One discrepant reading proves that all metal wires disobey Ohm's law.

Question 10 | 1 mark

special-39-04-q10

A 4Ω test wire must be replaced by a 1Ω wire of the same material and length. Which change to the conducting diameter achieves this? [1 mark]

- A. Halve the diameter.
- B. Keep the diameter unchanged.
- C. Multiply the diameter by four.
- D. Double the diameter.

Question 11 | 1 mark

special-39-04-q11

Two coils have resistances 4Ω and 8Ω . A student states, "The 8Ω coil always produces more heat in one minute." All currents and applied voltages in the choices are nonzero. Which test condition makes this claim false? [1 mark]

- A. The same current is maintained in both coils.
- B. Each coil is connected separately across the same ideal voltage.
- C. Both carry 1 A for one minute.
- D. The same charge crosses each in one minute at steady current.

Question 12 | 1 mark

special-39-04-q12

A device consumes 0.06 kWh of electrical energy. What is this energy in joules? [1 mark]

- A. 216 J

- B. 3600 J
- C. 216000 J
- D. 60000 J

Question 13 | 1 mark

special-39-04-q13

Two identical uniform wires each have length L and resistance R . Design E cuts one wire into two equal pieces and connects those pieces in parallel. Design U cuts the other wire into lengths $L/4$ and $3L/4$ and connects those pieces in parallel. Both designs use the entire wire, with negligible cutting loss, and are tested separately across the same nonzero ideal voltage. Which design draws more source current? [1 mark]

- A. U, because its equivalent resistance is $3R/16$, smaller than E's $R/4$.
- B. E, because equal branches always draw more total current.
- C. Both draw the same current because the total wire length is equal.
- D. U, because its equivalent resistance is $3R/4$, larger than E's $R/4$.

Question 14 | 1 mark

special-39-04-q14

A student compares two samples of equal length but different materials and different conducting areas. One sample has a larger measured resistance. Which extra measurement would allow a fair resistivity comparison? [1 mark]

- A. Only the colour of each sample.
- B. Only the mass of the plastic covering.
- C. Only the maximum current rating of the source.
- D. The conducting cross-sectional area of each sample.

Question 15 | 1 mark

special-39-04-q15

A $10\ \Omega$ resistor is operated at 2 V. A designer then uses a $20\ \Omega$ resistor at 4 V. Compared with the first arrangement, the second has: [1 mark]

- A. half the current and the same power.
- B. the same current and twice the power.
- C. twice the current and four times the power.
- D. the same current and the same power.

Question 16 | 1 mark

special-39-04-q16

A metal strip is bent into a loose coil without stretching it; adjacent turns do not touch. Which test result would be expected before and after bending, at the same temperature? [1 mark]

- A. Resistance increases because the path looks curved.
- B. Resistivity decreases because the strip occupies less space.
- C. The same end-to-end resistance, because material, path length and cross-section are unchanged.
- D. Zero resistance because turns are close together.

Question 17 | 1 mark

special-39-04-q17

A sample is known to be one of three chemically different candidate materials K, M or N. At the test temperature, K and M each have resistivity $8.0 \times 10^{-7} \, \Omega \, \text{m}$, while N has resistivity $1.2 \times 10^{-6} \, \Omega \, \text{m}$. The sample's resistivity is measured exactly as $8.0 \times 10^{-7} \, \Omega \, \text{m}$.

Assertion (A): This measurement rules out N and uniquely identifies the sample as K.

Reason (R): The supplied candidate data include two chemically different materials with the same resistivity at the test temperature. [1 mark]

- A. Both A and R are true, and R correctly explains A.
- B. Both A and R are true, but R does not correctly explain A.
- C. A is true, but R is false.
- D. A is false, but R is true.

Question 18 | 1 mark

special-39-04-q18

Assertion (A): A voltmeter is connected in parallel across a component.

Reason (R): One ampere means one coulomb of charge passes a cross-section each second. [1 mark]

- A. Both A and R are true, and R correctly explains A.
- B. Both A and R are true, but R does not correctly explain A.
- C. A is true, but R is false.
- D. A is false, but R is true.

Question 19 | 1 mark

special-39-04-q19

Assertion (A): For wires of equal material and length, doubling the conducting diameter halves the resistance.

Reason (R): The conducting cross-sectional area is proportional to the square of the diameter. [1 mark]

- A. Both A and R are true, and R correctly explains A.
- B. Both A and R are true, but R does not correctly explain A.
- C. A is true, but R is false.
- D. A is false, but R is true.

Question 20 | 1 mark

special-39-04-q20

Assertion (A): Connecting a second positive, finite resistance in parallel with a resistor across an ideal fixed-voltage source increases the source current.

Reason (R): The voltage across the original resistor increases when the branch is added. [1 mark]

- A. Both A and R are true, and R correctly explains A.
- B. Both A and R are true, but R does not correctly explain A.
- C. A is true, but R is false.
- D. A is false, but R is true.

Section B

Question 21 | 2 marks

special-39-04-q21

Two conductors are tested separately. In X, 72 C crosses a section in 3 minutes. In Y, 90 C crosses a section in 5 minutes.

- (i) Calculate both average currents. [1 mark]
- (ii) A learner ranks Y as having the larger current because more charge passed through it. Correct the ranking with a reason. [1 mark]

Question 22 | 2 marks

special-39-04-q22

Two sources deliver equal electrical energy, 180 J each, to resistors. Source U moves 30 C; source V moves 15 C.

- (i) Determine which source has greater potential difference and by what factor. [1 mark]
- (ii) Will equal amounts of charge receive equal energy from these two sources? Justify. [1 mark]

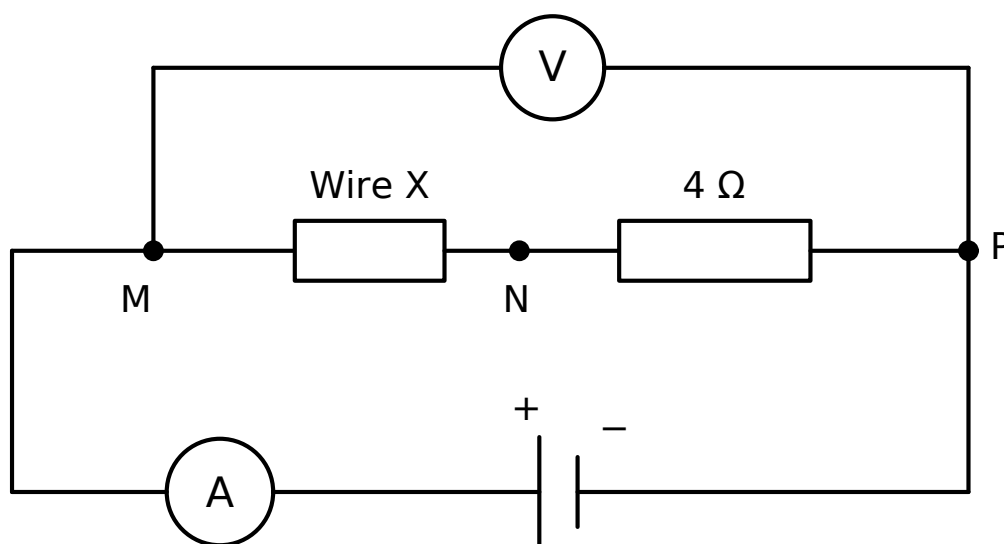
Question 23 | 2 marks

special-39-04-q23

A technician intends to measure only the resistance of wire X. The supplied diagram shows the actual meter connections. The ammeter reads 0.30 A and the voltmeter reads 3.0 V.

- (i) Explain why using 3.0/0.30 does not give the resistance of X, and state the correct terminals for the voltmeter. [1 mark]
- (ii) Find the resistance of X using the diagram and readings. [1 mark]

Actual measurement arrangement



Ideal source and meters

Question 24 | 2 marks

special-39-04-q24

For a uniform conductor, write the relation connecting resistance R , resistivity ρ , length L and conducting cross-sectional area A .

- (i) Use it to express ρ . [1 mark]
- (ii) State what physical property ρ describes and its SI unit. [1 mark]

Question 25 | 2 marks

special-39-04-q25

A constant 0.50 A flows through an $8\ \Omega$ resistor for 120 s.

- (i) Calculate the heat generated. [1 mark]
- (ii) The current is doubled but the operating time is halved. Without assuming unchanged voltage, determine the new heat and explain the factor of change. [1 mark]

Question 26 | 2 marks

special-39-04-q26

In this test only, the wire temperature is allowed to change. A uniform wire draws 0.40 A at 2.0 V when cool and 0.32 A at 2.0 V after warming. Treat its length and conducting area as unchanged.

- (i) Find the ratio of its warm resistivity to its cool resistivity. [1 mark]
- (ii) A learner says the rise in resistance proves that the wire has become a different material. Explain why this is not established, and propose a useful repeat measurement. [1 mark]

Section C

Question 27 | 3 marks

special-39-04-q27

The diagram gives measured resistances and dimensions of three uniform samples P, Q and T at one temperature.

- (i) Which pair has equal resistivity? Show a comparison using the labelled values. [1 mark]
 (ii) Find ρ_T/ρ_P . [1 mark]
 (iii) A student says, "Q must be made of the highest-resistivity material because it has the greatest resistance." Explain the error using this evidence. [1 mark]

Sample dimensions and measured resistance

P	$L = 1.0 \text{ m}$		$R = 4 \Omega$
	Conducting area = 0.50 mm^2		
Q	$L = 2.0 \text{ m}$		$R = 8 \Omega$
	Conducting area = 0.50 mm^2		
T	$L = 1.0 \text{ m}$		$R = 4 \Omega$
	Conducting area = 1.00 mm^2		

All samples measured at the same temperature.

Question 28 | 3 marks

special-39-04-q28

A spool is sampled at the same temperature and conducting area. At 2.0 V, lengths 0.50 m, 1.00 m and 2.00 m draw 0.40 A, 0.20 A and 0.10 A respectively.

- (i) Calculate R/L for all three samples and state whether they support a common resistance-per-length model. [1 mark]
 (ii) A sensor test requires current from 0.12 A to 0.15 A inclusive at 2.0 V. Choose a length from 1.0 m, 1.5 m and 2.0 m, showing why the others fail. [2 marks]

Question 29 | 3 marks

special-39-04-q29

Two unknown resistors X and Y are in series across 6.0 V. The steady current is 0.50 A; the measured voltage across X is 2.0 V.

- (i) Find the resistance of X. [1 mark]
- (ii) Infer the voltage and resistance of Y. [1 mark]
- (iii) A student predicts equal heat in X and Y over the same time because their currents are equal. Evaluate this prediction. [1 mark]

Question 30 | 3 marks

special-39-04-q30

A single continuous wire has two uniform sections made of the same material, joined end to end with no branch and negligible junction resistance. Section X is 0.40 m long with area 0.20 mm²; section Y is 0.80 m long with area 0.40 mm². The operating resistivity of both is $1.0 \times 10^{-6} \Omega \text{ m}$ and is treated as constant. For part (iii), override the general constant-temperature and neglected-loss assumptions: initial temperatures and heat-loss conditions are unspecified, and temperatures may change. Continue to treat the stated electrical resistivity as constant; no thermal calculation is required. A steady current of 0.50 A flows through the wire for 40 s.

- (i) Find the resistance and total heat generated in each section. [1 mark]
- (ii) Power generated per metre means a section's electrical power divided by its length. A designer wants the larger heat-generation rate per metre. Choose X or Y and justify numerically, explaining why the current is not smaller in the narrower section. [1 mark]
- (iii) A learner claims that your choice in (ii) must reach the higher temperature. Is that conclusion established by these electrical data alone? Explain without using a temperature-rise formula. [1 mark]

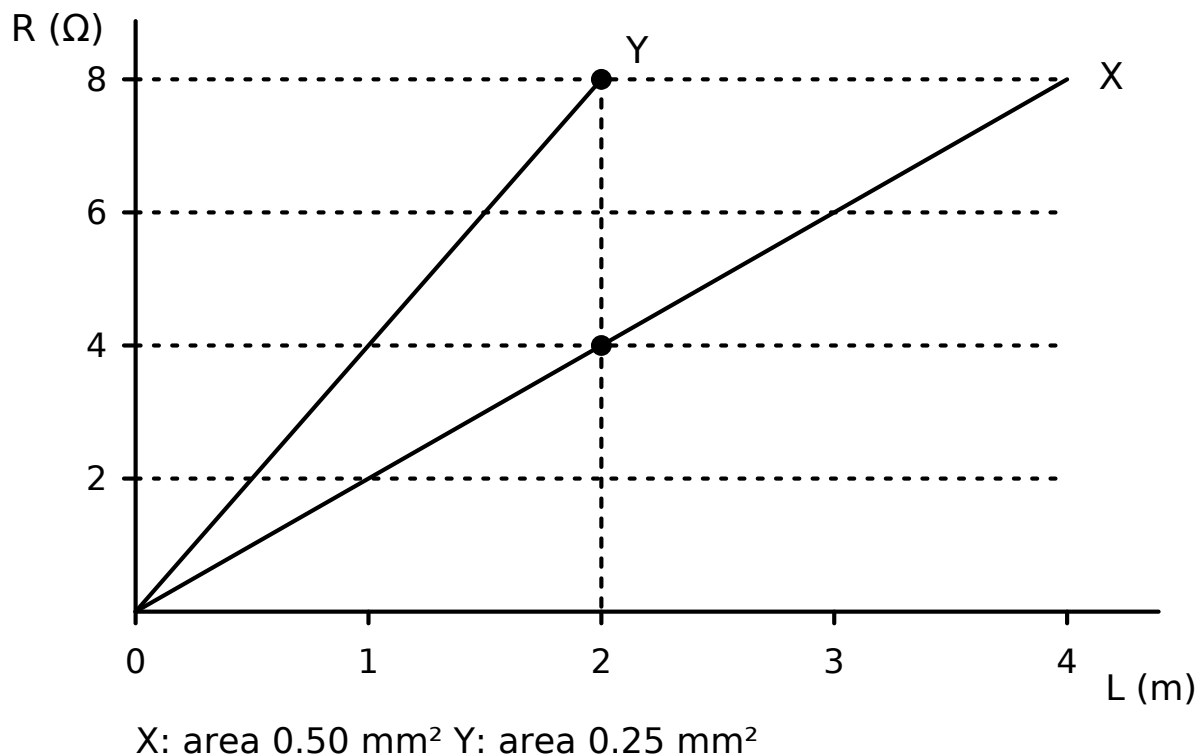
Question 31 | 3 marks

special-39-04-q31

The plotted lines show resistance against length for two uniform wire stocks. Stock X has conducting area 0.50 mm²; stock Y has conducting area 0.25 mm².

- (i) Read each resistance-per-metre value from the graph. [1 mark]
- (ii) Compare their resistivities. [1 mark]
- (iii) A 6 Ω resistor must fit a path of at most 2.0 m. It must be a single piece with the stock area unchanged. Which stock can meet this requirement, and what length should be cut? [1 mark]

Resistance measured at constant temperature



Question 32 | 3 marks

special-39-04-q32

A workshop's cumulative electrical-energy meter reads 1250 kWh at the start of a billing period and 1268 kWh at its end. All the metered consumption occurred during a total of 30 energised hours; there was no consumption outside those hours. The tariff is ₹4 per kWh for the first 10 kWh consumed in this period and ₹7 per kWh for only the consumption above 10 kWh; ignore fixed charges and taxes. A proposed replacement is independently confirmed to give the same required useful output while reducing the period's electrical consumption by 6 kWh.

- Find the consumed electrical energy and the average input power during the 30 energised hours. [1 mark]
- Calculate the original bill using the stated tariff. [1 mark]
- A manager predicts a saving of only ₹24 by pricing all 6 saved kWh at ₹4. Calculate the actual bill reduction and assess this prediction. [1 mark]

Question 33 | 3 marks

special-39-04-q33

A heater must have resistance 20Ω , length 1.0 m and conducting area 0.050 mm^2 . Three hypothetical materials are offered: K has resistivity $1.0 \times 10^{-6} \Omega \text{ m}$ and withstands the operating temperature without rapid oxidation; L has resistivity $2.0 \times 10^{-8} \Omega \text{ m}$ and also withstands it; M has resistivity $1.0 \times 10^{-6} \Omega \text{ m}$ but rapidly oxidises at that temperature. Use the supplied operating resistivities.

- Determine which materials meet the electrical resistance requirement. [1 mark]
- Choose the suitable material when durability is also required, and explain why each other material fails a requirement. [1 mark]

(iii) If the length of L stays 1.0 m, calculate the conducting area needed to reach $20\ \Omega$ and explain why this cannot satisfy the original fixed-geometry requirement. [1 mark]

Section D

Question 34 | 5 marks

special-39-04-q34

A manufacturer has the original uniform wire shown in the figure. The full wire must be redrawn into one uniform wire, with no material removed or added and no change in resistivity. A $1.00\ \Omega$ component is required. It must also satisfy the maximum length and minimum conducting area stated on the drawing.

- (i) Calculate the original conducting-material volume and explain the relation between the new area A' and length L' . [1 mark]
- (ii) Determine the length and area needed for $1.00\ \Omega$, using conservation of volume rather than assuming fixed area. [2 marks]
- (iii) Does the proposed wire satisfy both manufacturing limits? Show the checks. [1 mark]
- (iv) Compare its power with that of the original wire when each is separately tested across 2.0 V . [1 mark]

Original wire and manufacturing limits

Length = 0.80 m



Conducting area = 0.50 mm^2

Measured resistance = $0.25\ \Omega$

Limits for the new uniform wire:

Maximum length: 1.70 m

Minimum conducting area: 0.20 mm^2

Use all the original conducting material; no loss.

Question 35 | 5 marks

special-39-04-q35

A laboratory receives wires A and B. A has length 1.0 m and diameter 0.20 mm ; B has length 2.0 m and diameter 0.40 mm . Separate tests at 1.2 V give currents 0.30 A in A and 0.60 A in B.

- (i) Calculate the measured resistance of each wire. [1 mark]
- (ii) Without needing π , determine the ratio of their resistivities. Explain why B's lower resistance does not establish a better-conducting material. [2 marks]
- (iii) The lab needs a $4.0\ \Omega$ resistor from stock of B's material and diameter, with at most 3.0 m

allowed. Determine whether it is possible with this stock unchanged. [1 mark]

(iv) If exactly 2.0 m must instead be used, specify the conducting area as a fraction of B's area that would give 4.0 Ω . [1 mark]

Question 36 | 5 marks

special-39-04-q36

Two candidate uniform resistive pads are to be formed separately from equal volumes, $4.0 \times 10^{-7} \text{ m}^3$, of the same alloy. Its operating resistivity is $1.0 \times 10^{-6} \Omega \text{ m}$. Candidate U has wire length 1.0 m; candidate V has wire length 2.0 m. Each candidate uses its entire allotted volume, without loss. Either pad would be connected directly across an ideal 10 V source. The source must supply no more than 2.0 A. A job requires 6000 J of useful heat within 700 s, and all electrical energy reaches the load.

(i) Find the conducting area and resistance of each candidate. [2 marks]

(ii) Find their currents and decide which meets the source-current limit. [1 mark]

(iii) Check whether the current-compatible candidate meets the time limit. [1 mark]

(iv) A seller claims the rejected, higher-power pad would use less electrical energy for the same job. Assess this claim under the stated assumptions. [1 mark]

Section E

Question 37 | 4 marks

special-39-04-q37

A class tests different lengths from one wire stock at constant temperature using two clamps. Unlike the general ideal-contact assumption, the two clamp contacts together contribute an unknown series resistance C . In the first two trials the stock has a uniform conducting area and material, and C stays unchanged. The measured total resistance (wire plus both contacts) is $4.0\ \Omega$ for $0.50\ \text{m}$ and $10.0\ \Omega$ for $1.50\ \text{m}$. Connecting leads have negligible resistance. Treat the readings as exact. For these trials write total resistance as $kL + C$, where k is the wire resistance per metre.

- (i) Use the difference between the trials to infer k , then find C . Explain why simply dividing either total resistance by its length would give a misleading k . [2 marks]
- (ii) Predict the total resistance for $2.00\ \text{m}$ if the uniform-stock and unchanged-contact model continues to apply. [1 mark]
- (iii) The $2.00\ \text{m}$ trial instead gives $15.0\ \Omega$. Consider only two explanations: H1, the contacts now add $2.0\ \Omega$ extra at every length while the wire remains uniform; H2, the contacts are unchanged but the added segment from $1.50\ \text{m}$ to $2.00\ \text{m}$ has extra resistance. The contact condition can be held unchanged while returning to the same previously tested $0.50\ \text{m}$ or $1.50\ \text{m}$ portion, excluding the added $1.50\text{--}2.00\ \text{m}$ segment. Specify one shorter-length repeat and its predicted reading under each explanation. Does the single $15.0\ \Omega$ reading alone select between H1 and H2? [1 mark]

Question 38 | 4 marks

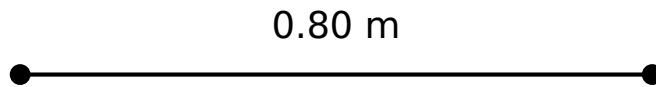
special-39-04-q38

A design team must make one continuous resistive track by cutting a uniform stock wire to a chosen path. The layout drawing supplies the two allowed paths and the minimum conducting area. It is not a circuit diagram: the two paths are alternatives and are never connected together. The wire's operating resistivity is $1.0 \times 10^{-6}\ \Omega\ \text{m}$. The track is connected directly across $6.0\ \text{V}$, must draw no more than $1.2\ \text{A}$, and must deliver at least $360\ \text{J}$ in $60\ \text{s}$. All electrical energy becomes useful heat. Only the two shown path lengths may be used; the conducting area may be chosen but must satisfy the drawing's minimum.

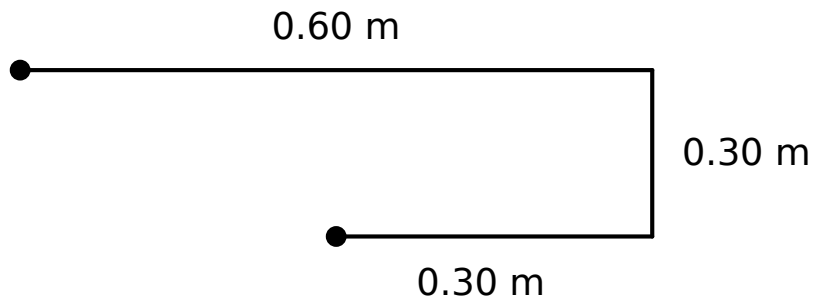
- (i) At the minimum permitted area, find the resistance of each path and reject any path that already draws too much current. [1 mark]
- (ii) Can a permitted increase in conducting area rescue a path rejected in (i)? Explain. [1 mark]
- (iii) For the remaining path at the minimum area, check both the source-current and heating requirements, and state a feasible design. [2 marks]

Alternative track paths (not connected together)

Path S



Path T



Minimum conducting area for either path:
 0.20 mm^2

Ends are terminals; dimensions give actual wire length.

Question 39 | 4 marks

special-39-04-q39

A 12 V laboratory station has two constant-resistance devices: a warming pad rated 12 V, 12 W and an indicator rated 12 V, 3 W. They must each receive 12 V when switched on. Each device has its own switch. In a daily session the pad runs for 20 minutes and the indicator for 1 hour; their operating periods may overlap. Assume an ideal 12 V supply, negligible switch/wire resistance, and no standby consumption.

- (i) Choose series or parallel connection and explain how it meets both the voltage and independent-switching requirements. [1 mark]
- (ii) Find the total source current when both devices are on. [1 mark]
- (iii) A proposed operating schedule halves the pad's daily on-time but leaves the indicator unchanged. Calculate the daily energy saved and the percentage reduction in the total daily electrical energy. [2 marks]

Detailed solutions and marking guidance

Original model answers, numbered to match the question paper. These are not an official marking scheme.

Solution 1 — special-39-04-q01

Correct option: B

B. Current is the rate of flow of charge across a cross-section. An ammeter is connected in series with the resistor so that the current being measured passes through the meter. The other choices confuse current with energy or voltage, or put the ammeter across the resistor. [1 mark]

Mark allocation

B. Current is the rate of flow of charge across a cross-section. An ammeter is connected in series with the resistor so that the current being measured passes through the meter. The other choices confuse current with energy or voltage, or put the ammeter across the resistor. [1 marks]

Solution 2 — special-39-04-q02

Correct option: C

C. Electrical work is voltage multiplied by charge. Thus $W_Y = (3V)(Q/2) = (3/2)VQ = 3W/2$. Multiplying the voltage ratio 3 by the charge ratio $1/2$ gives $3/2$, not 6 or $1/6$. [1 mark]

Mark allocation

C. Electrical work is voltage multiplied by charge. Thus $W_Y = (3V)(Q/2) = (3/2)VQ = 3W/2$. Multiplying the voltage ratio 3 by the charge ratio $1/2$ gives $3/2$, not 6 or $1/6$. [1 marks]

Solution 3 — special-39-04-q03

Correct option: A

A. At 4 V the graph gives $I_P = 0.8$ A and $I_Q = 0.4$ A, so $R_P = 5 \Omega$ and $R_Q = 10 \Omega$. Resistivity is RA/L , hence $\rho_Q/\rho_P = (10/5)(A/A)(L/4L) = 1/2$. Thus P has twice Q's resistivity even though P has the smaller measured resistance. Ignoring the fourfold length would reverse this material ranking; slope on these axes is I/V , not R . [1 mark]

Mark allocation

A. At 4 V the graph gives $I_P = 0.8$ A and $I_Q = 0.4$ A, so $R_P = 5 \Omega$ and $R_Q = 10 \Omega$. Resistivity is RA/L , hence $\rho_Q/\rho_P = (10/5)(A/A)(L/4L) = 1/2$. Thus P has twice Q's resistivity even though P has the smaller measured resistance. Ignoring the fourfold length would reverse this material ranking; slope on these axes is I/V , not R . [1 marks]

Solution 4 — special-39-04-q04

Correct option: C

C. $\rho = RA/L$. For Y, the factors of 2 in area and length cancel; both give $\rho = 8A/L$. Equal resistivity is not a unique chemical identification. [1 mark for the correct option.]

Mark allocation

Select C; $\rho = RA/L$. For Y, the factors of 2 in area and length cancel; both give $\rho = 8A/L$. Equal resistivity is not a unique chemical identification. [1 marks]

Solution 5 — special-39-04-q05

Correct option: A

A. $R = \rho L/A$ uses the conducting metal area, not the area enclosed by the insulation. None of ρ , L or conducting A changes. [1 mark for the correct option.]

Mark allocation

Select A; $R = \rho L/A$ uses the conducting metal area, not the area enclosed by the insulation. None of ρ , L or conducting A changes. [1 marks]

Solution 6 — special-39-04-q06

Correct option: D

D. The two $6\ \Omega$ resistors share the same two nodes, so their parallel equivalent is $3\ \Omega$. With the $3\ \Omega$ series resistor, $R_{\text{total}} = 6\ \Omega$ and $I_{\text{source}} = 9/6 = 1.5\ \text{A}$. Identical parallel branches divide this equally: $0.75\ \text{A}$ each. [1 mark for the correct option.]

Mark allocation

Select D; The two $6\ \Omega$ resistors share the same two nodes, so their parallel equivalent is $3\ \Omega$. With the $3\ \Omega$ series resistor, $R_{\text{total}} = 6\ \Omega$ and $I_{\text{source}} = 9/6 = 1.5\ \text{A}$. Identical parallel branches divide this equally: $0.75\ \text{A}$ each. [1 marks]

Solution 7 — special-39-04-q07

Correct option: B

B. The steady current is the same on both sides: $I = 18/6 = 3\ \text{A}$. Thus $18\ \text{C}$ also leaves in $6\ \text{s}$. Heating transfers energy, not charge into heat. [1 mark for the correct option.]

Mark allocation

Select B; The steady current is the same on both sides: $I = 18/6 = 3\ \text{A}$. Thus $18\ \text{C}$ also leaves in $6\ \text{s}$. Heating transfers energy, not charge into heat. [1 marks]

Solution 8 — special-39-04-q08

Correct option: C

C. Since $1\ \text{mm} = 10^{-3}\ \text{m}$, $1\ \text{mm}^2 = (10^{-3}\ \text{m})^2 = 10^{-6}\ \text{m}^2$. Therefore $0.35\ \text{mm}^2 = 0.35 \times 10^{-6}\ \text{m}^2 = 3.5 \times 10^{-7}\ \text{m}^2$. The conversion factor must be squared because the quantity is an area. [1 mark]

Mark allocation

C. Since $1\ \text{mm} = 10^{-3}\ \text{m}$, $1\ \text{mm}^2 = (10^{-3}\ \text{m})^2 = 10^{-6}\ \text{m}^2$. Therefore $0.35\ \text{mm}^2 = 0.35 \times 10^{-6}\ \text{m}^2 = 3.5 \times 10^{-7}\ \text{m}^2$. The conversion factor must be squared because the quantity is an area. [1 marks]

Solution 9 — special-39-04-q09

Correct option: A

A. The first three ratios V/I equal $6\ \Omega$; the original final row gives $8\ \Omega$, while the repeat gives $6\ \Omega$. The repeat supports investigating the outlier, not declaring a universal failure or silently discarding data. [1 mark for the correct option.]

Mark allocation

Select A; The first three ratios V/I equal $6\ \Omega$; the original final row gives $8\ \Omega$, while the repeat gives $6\ \Omega$. The repeat supports investigating the outlier, not declaring a universal failure or silently discarding data. [1 marks]

Solution 10 — special-39-04-q10

Correct option: D

D. At fixed p and L , $R \propto 1/A$ and $A \propto d^2$. Doubling d gives four times the area and one-quarter the resistance: $1\ \Omega$. [1 mark for the correct option.]

Mark allocation

Select D; At fixed p and L , $R \propto 1/A$ and $A \propto d^2$. Doubling d gives four times the area and one-quarter the resistance: $1\ \Omega$. [1 marks]

Solution 11 — special-39-04-q11

Correct option: B

B. At equal voltage and equal time, $H = V^2t/R$, so the $8\ \Omega$ coil produces half the heat of the $4\ \Omega$ coil. The other choices imply equal current and time, when $H = I^2Rt$ gives more heat in $8\ \Omega$. [1 mark for the correct option.]

Mark allocation

Select B; At equal voltage and equal time, $H = V^2t/R$, so the $8\ \Omega$ coil produces half the heat of the $4\ \Omega$ coil. The other choices imply equal current and time, when $H = I^2Rt$ gives more heat in $8\ \Omega$. [1 marks]

Solution 12 — special-39-04-q12

Correct option: C

C. $1\ \text{kWh} = 3.6 \times 10^6\ \text{J}$, hence $0.06\ \text{kWh} = 216000\ \text{J}$. [1 mark for the correct option.]

Mark allocation

Select C; $1\ \text{kWh} = 3.6 \times 10^6\ \text{J}$, hence $0.06\ \text{kWh} = 216000\ \text{J}$. [1 marks]

Solution 13 — special-39-04-q13

Correct option: A

A. E has two $R/2$ pieces in parallel, giving $R_E = R/4$. U has $R/4$ and $3R/4$ in parallel, giving $R_U = (R/4)(3R/4)/R = 3R/16$. At the same voltage, $I_U/I_E = R_E/R_U = 4/3$, so U draws more source current. Equal total length does not fix equivalent resistance after reconnection. [1 mark for the correct option.]

Mark allocation

A. E has two $R/2$ pieces in parallel, giving $R_E = R/4$. U has $R/4$ and $3R/4$ in parallel, giving $R_U = (R/4)(3R/4)/R = 3R/16$. At the same voltage, $I_U/I_E = R_E/R_U = 4/3$, so U draws more source

current. Equal total length does not fix equivalent resistance after reconnection. [1 mark for the correct option.] [1 marks]

Solution 14 — special-39-04-q14

Correct option: D

D. Measure each conducting area, then use $\rho = RA/L$ at the same temperature. Larger R alone cannot rank material resistivity when A differs. [1 mark for the correct option.]

Mark allocation

Select D; Measure each conducting area, then use $\rho = RA/L$ at the same temperature. Larger R alone cannot rank material resistivity when A differs. [1 marks]

Solution 15 — special-39-04-q15

Correct option: B

B. The currents are $2/10 = 0.20$ A and $4/20 = 0.20$ A. $P = VI$ gives 0.40 W and 0.80 W, so power doubles despite unchanged current. [1 mark for the correct option.]

Mark allocation

Select B; The currents are $2/10 = 0.20$ A and $4/20 = 0.20$ A. $P = VI$ gives 0.40 W and 0.80 W, so power doubles despite unchanged current. [1 marks]

Solution 16 — special-39-04-q16

Correct option: C

C. $R = \rho L/A$ is unchanged: bending without stretching or contacts changes neither the actual conducting length nor its area or material. [1 mark for the correct option.]

Mark allocation

Select C; $R = \rho L/A$ is unchanged: bending without stretching or contacts changes neither the actual conducting length nor its area or material. [1 marks]

Solution 17 — special-39-04-q17

Correct option: D

D. A is false: the measurement excludes N but cannot distinguish K from M, so it does not uniquely identify K. R is true: the stated K and M values are equal despite their different compositions. A matching resistivity can leave more than one candidate; an unequal value at the controlled temperature can exclude a candidate. [1 mark]

Mark allocation

D. A is false: the measurement excludes N but cannot distinguish K from M, so it does not uniquely identify K. R is true: the stated K and M values are equal despite their different compositions. A matching resistivity can leave more than one candidate; an unequal value at the controlled temperature can exclude a candidate. [1 marks]

Solution 18 — special-39-04-q18

Correct option: B

B. Both statements are true. The reason defines current, but does not explain voltmeter connection; a voltmeter compares potentials at the two component terminals. [1 mark for the correct option.]

Mark allocation

Select B; Both statements are true. The reason defines current, but does not explain voltmeter connection; a voltmeter compares potentials at the two component terminals. [1 marks]

Solution 19 — special-39-04-q19

Correct option: D

D. A is false: doubling diameter multiplies area by four and reduces resistance to one-quarter, not one-half. R is true and identifies the missed squared relationship. [1 mark for the correct option.]

Mark allocation

Select D; A is false: doubling diameter multiplies area by four and reduces resistance to one-quarter, not one-half. R is true and identifies the missed squared relationship. [1 marks]

Solution 20 — special-39-04-q20

Correct option: C

C. A is true: the new branch draws additional current while the original branch current remains V/R . R is false because the ideal source maintains the same voltage across both branches. [1 mark for the correct option.]

Mark allocation

Select C; A is true: the new branch draws additional current while the original branch current remains V/R . R is false because the ideal source maintains the same voltage across both branches. [1 marks]

Solution 21 — special-39-04-q21

(i) $I_X = 72/180 = 0.40$ A and $I_Y = 90/300 = 0.30$ A (0.5 mark each). [1 mark]

(ii) X has the larger average current. Current compares charge per unit time, not total charge measured over different times. [1 mark]

Mark allocation

(i) $I_X = 72/180 = 0.40$ A and $I_Y = 90/300 = 0.30$ A (0.5 mark each). [1 marks]

(ii) X has the larger average current. Current compares charge per unit time, not total charge measured over different times. [1 marks]

Solution 22 — special-39-04-q22

(i) $V_U = 180/30 = 6$ V; $V_V = 180/15 = 12$ V. Source V has twice the potential difference despite equal total energy delivered in the stated trials. [1 mark]

(ii) No. $W = VQ$, so for any same nonzero charge Q , source V transfers twice the energy transferred by U. Equal trial energies arose from different charges. [1 mark]

Mark allocation

(i) $V_U = 180/30 = 6 \text{ V}$; $V_V = 180/15 = 12 \text{ V}$. Source V has twice the potential difference despite equal total energy delivered in the stated trials. [1 marks]

(ii) No. $W = VQ$, so for any same nonzero charge Q , source V transfers twice the energy transferred by U. Equal trial energies arose from different charges. [1 marks]

Solution 23 — special-39-04-q23

(i) The voltmeter spans M to P, including X and the 4Ω resistor. It should span M to N to measure X alone; its present reading divided by current is the combined resistance. [1 mark]

(ii) $R_{MP} = 3.0/0.30 = 10 \Omega$. Since the same current passes through X and 4Ω , $R_X = 10 - 4 = 6 \Omega$. Equivalently $V_X = 3.0 - 0.30 \times 4 = 1.8 \text{ V}$ and $R_X = 1.8/0.30 = 6 \Omega$. [1 mark]

Mark allocation

(i) The voltmeter spans M to P, including X and the 4Ω resistor. It should span M to N to measure X alone; its present reading divided by current is the combined resistance. [1 marks]

(ii) $R_{MP} = 3.0/0.30 = 10 \Omega$. Since the same current passes through X and 4Ω , $R_X = 10 - 4 = 6 \Omega$. Equivalently $V_X = 3.0 - 0.30 \times 4 = 1.8 \text{ V}$ and $R_X = 1.8/0.30 = 6 \Omega$. [1 marks]

Solution 24 — special-39-04-q24

(i) $R = \rho L/A$, so $\rho = RA/L$. [1 mark]

(ii) Resistivity characterises the material's opposition to current independently of the specimen dimensions at a specified temperature (0.5). Its SI unit is $\Omega \text{ m}$ (0.5). [1 mark]

Mark allocation

(i) $R = \rho L/A$, so $\rho = RA/L$. [1 marks]

(ii) Resistivity characterises the material's opposition to current independently of the specimen dimensions at a specified temperature (0.5). Its SI unit is $\Omega \text{ m}$ (0.5). [1 marks]

Solution 25 — special-39-04-q25

(i) $H = I^2 R t = (0.50)^2 \times 8 \times 120 = 240 \text{ J}$. [1 mark]

(ii) $H'/H = (2I)^2(R)(t/2)/(I^2 R t) = 2$, so $H' = 480 \text{ J}$. Squaring the current gives a factor of four, then halving time gives an overall factor of two. [1 mark]

Mark allocation

(i) $H = I^2 R t = (0.50)^2 \times 8 \times 120 = 240 \text{ J}$. [1 marks]

(ii) $H'/H = (2I)^2(R)(t/2)/(I^2 R t) = 2$, so $H' = 480 \text{ J}$. Squaring the current gives a factor of four, then halving time gives an overall factor of two. [1 marks]

Solution 26 — special-39-04-q26

(i) $R_{\text{cool}} = 2.0/0.40 = 5.0 \Omega$ and $R_{\text{warm}} = 2.0/0.32 = 6.25 \Omega$. Since L and A are unchanged, $\rho_{\text{warm}}/\rho_{\text{cool}} = R_{\text{warm}}/R_{\text{cool}} = 6.25/5.0 = 1.25$. [1 mark]

(ii) Resistivity can vary with temperature even without a change in material, so the two-temperature comparison does not establish a new composition (0.5 mark). Cool the wire back to the original temperature and repeat the same-voltage measurement; recovery of the original current would support a reversible temperature effect (0.5 mark). [1 mark]

Mark allocation

(i) $R_{\text{cool}} = 2.0/0.40 = 5.0 \, \Omega$ and $R_{\text{warm}} = 2.0/0.32 = 6.25 \, \Omega$. Since L and A are unchanged, $\rho_{\text{warm}}/\rho_{\text{cool}} = R_{\text{warm}}/R_{\text{cool}} = 6.25/5.0 = 1.25$. [1 marks]

(ii) Resistivity can vary with temperature even without a change in material, so the two-temperature comparison does not establish a new composition (0.5 mark). Cool the wire back to the original temperature and repeat the same-voltage measurement; recovery of the original current would support a reversible temperature effect (0.5 mark). [1 marks]

Solution 27 — special-39-04-q27

(i) Compare RA/L in common units. P gives $4 \times 0.50/1.0 = 2$ and Q gives $8 \times 0.50/2.0 = 2$ in $\Omega \text{ mm}^2/\text{m}$; hence P and Q have equal resistivity ($2 \times 10^{-6} \, \Omega \text{ m}$). [1 mark]

(ii) T gives $4 \times 1.00/1.0 = 4$ in the same units. Thus $\rho_T/\rho_P = 4/2 = 2$. [1 mark]

(iii) Q 's higher resistance than P is explained by its doubled length, not by higher resistivity. T has the greatest resistivity even though its measured resistance equals P 's, because T has twice P 's area at the same length. [1 mark]

Mark allocation

(i) Compare RA/L in common units. P gives $4 \times 0.50/1.0 = 2$ and Q gives $8 \times 0.50/2.0 = 2$ in $\Omega \text{ mm}^2/\text{m}$; hence P and Q have equal resistivity ($2 \times 10^{-6} \, \Omega \text{ m}$). [1 marks]

(ii) T gives $4 \times 1.00/1.0 = 4$ in the same units. Thus $\rho_T/\rho_P = 4/2 = 2$. [1 marks]

(iii) Q 's higher resistance than P is explained by its doubled length, not by higher resistivity. T has the greatest resistivity even though its measured resistance equals P 's, because T has twice P 's area at the same length. [1 marks]

Solution 28 — special-39-04-q28

(i) Resistances are 5, 10 and 20 Ω , respectively. Dividing by 0.50, 1.00 and 2.00 m gives 10 Ω/m in every case, supporting $R = (10 \, \Omega/\text{m})L$ for this spool under the test conditions. [1 mark]

(ii) The three candidate resistances are 10, 15 and 20 Ω . Their currents are 0.20 A, $2/15 \text{ A} \approx 0.133 \text{ A}$ and 0.10 A (1 mark). Choose 1.5 m: only 0.133 A is within 0.12–0.15 A; 1.0 m draws too much and 2.0 m too little (1 mark). [2 marks]

Mark allocation

(i) Resistances are 5, 10 and 20 Ω , respectively. Dividing by 0.50, 1.00 and 2.00 m gives 10 Ω/m in every case, supporting $R = (10 \, \Omega/\text{m})L$ for this spool under the test conditions. [1 marks]

(ii) The three candidate resistances are 10, 15 and 20 Ω . Their currents are 0.20 A, $2/15 \text{ A} \approx 0.133 \text{ A}$ and 0.10 A (1 mark). Choose 1.5 m: only 0.133 A is within 0.12–0.15 A; 1.0 m draws too much and 2.0 m too little (1 mark). [2 marks]

Solution 29 — special-39-04-q29

(i) $R_X = V_X/I = 2.0/0.50 = 4.0 \, \Omega$. [1 mark]

(ii) Series voltages add, so $V_Y = 6.0 - 2.0 = 4.0 \text{ V}$. $R_Y = 4.0/0.50 = 8.0 \, \Omega$. [1 mark]

(iii) The prediction is false. At common current and time, $H = I^2Rt$, so $H_Y/H_X = 8/4 = 2$; Y generates twice the heat. [1 mark]

Mark allocation

(i) $R_X = V_X/I = 2.0/0.50 = 4.0 \, \Omega$. [1 marks]

(ii) Series voltages add, so $V_Y = 6.0 - 2.0 = 4.0 \, \text{V}$. $R_Y = 4.0/0.50 = 8.0 \, \Omega$. [1 marks]

(iii) The prediction is false. At common current and time, $H = I^2Rt$, so $H_Y/H_X = 8/4 = 2$; Y generates twice the heat. [1 marks]

Solution 30 — special-39-04-q30

(i) $A_X = 2.0 \times 10^{-7} \, \text{m}^2$ and $A_Y = 4.0 \times 10^{-7} \, \text{m}^2$. $R_X = 1.0 \times 10^{-6} \times 0.40 / (2.0 \times 10^{-7}) = 2.0 \, \Omega$ and $R_Y = 1.0 \times 10^{-6} \times 0.80 / (4.0 \times 10^{-7}) = 2.0 \, \Omega$ (0.5 mark for both resistances). Each generates $H = (0.50)^2 \times 2.0 \times 40 = 20 \, \text{J}$ (0.5 mark for both heats). [1 mark]

(ii) There is one steady unbranched path, so the current is 0.50 A in both sections; charge does not disappear or accumulate at the narrowing. Each generates $P = I^2R = 0.50 \, \text{W}$, but $P_X/L_X = 0.50/0.40 = 1.25 \, \text{W/m}$ and $P_Y/L_Y = 0.50/0.80 = 0.625 \, \text{W/m}$. Choose X: it has twice the heat-generation rate per metre, although the sections generate equal total heat. [1 mark]

(iii) No. A larger electrical heat-generation rate per metre does not by itself determine the temperature attained. Heat loss/cooling and the amount of material being warmed also matter. Those thermal conditions are not specified; the calculation establishes generated energy and power per metre, not a temperature ranking. No temperature change in resistivity is used in this model. [1 mark]

Mark allocation

(i) $A_X = 2.0 \times 10^{-7} \, \text{m}^2$ and $A_Y = 4.0 \times 10^{-7} \, \text{m}^2$. $R_X = 1.0 \times 10^{-6} \times 0.40 / (2.0 \times 10^{-7}) = 2.0 \, \Omega$ and $R_Y = 1.0 \times 10^{-6} \times 0.80 / (4.0 \times 10^{-7}) = 2.0 \, \Omega$ (0.5 mark for both resistances). Each generates $H = (0.50)^2 \times 2.0 \times 40 = 20 \, \text{J}$ (0.5 mark for both heats). [1 marks]

(ii) There is one steady unbranched path, so the current is 0.50 A in both sections; charge does not disappear or accumulate at the narrowing. Each generates $P = I^2R = 0.50 \, \text{W}$, but $P_X/L_X = 0.50/0.40 = 1.25 \, \text{W/m}$ and $P_Y/L_Y = 0.50/0.80 = 0.625 \, \text{W/m}$. Choose X: it has twice the heat-generation rate per metre, although the sections generate equal total heat. [1 marks]

(iii) No. A larger electrical heat-generation rate per metre does not by itself determine the temperature attained. Heat loss/cooling and the amount of material being warmed also matter. Those thermal conditions are not specified; the calculation establishes generated energy and power per metre, not a temperature ranking. No temperature change in resistivity is used in this model. [1 marks]

Solution 31 — special-39-04-q31

(i) At 2 m the graph gives 4 Ω for X and 8 Ω for Y. Thus R/L is 2 Ω/m for X and 4 Ω/m for Y. [1 mark]

(ii) $\rho = (R/L)A$. For X, $\rho_X = 2 \times 0.50 \times 10^{-6} = 1.0 \times 10^{-6} \, \Omega \, \text{m}$. For Y, $\rho_Y = 4 \times 0.25 \times 10^{-6} = 1.0 \times 10^{-6} \, \Omega \, \text{m}$. The stocks have equal resistivity despite unequal graph slopes. [1 mark]

(iii) X requires $L = 6/2 = 3 \, \text{m}$, exceeding the limit. Y requires $L = 6/4 = 1.5 \, \text{m}$, which fits. Choose 1.5 m of Y. [1 mark]

Mark allocation

(i) At 2 m the graph gives 4 Ω for X and 8 Ω for Y. Thus R/L is 2 Ω/m for X and 4 Ω/m for Y. [1 marks]

(ii) $\rho = (R/L)A$. For X, $\rho_X = 2 \times 0.50 \times 10^{-6} = 1.0 \times 10^{-6} \Omega \text{ m}$. For Y, $\rho_Y = 4 \times 0.25 \times 10^{-6} = 1.0 \times 10^{-6} \Omega \text{ m}$. The stocks have equal resistivity despite unequal graph slopes. [1 marks]

(iii) X requires $L = 6/2 = 3 \text{ m}$, exceeding the limit. Y requires $L = 6/4 = 1.5 \text{ m}$, which fits. Choose 1.5 m of Y. [1 marks]

Solution 32 — special-39-04-q32

(i) Consumed energy is the increase in the cumulative reading: $1268 - 1250 = 18 \text{ kWh}$ (0.5 mark). Average input power during energised hours = $18 \text{ kWh}/30 \text{ h} = 0.60 \text{ kW} = 600 \text{ W}$ (0.5 mark). [1 mark]

(ii) Only the first 10 kWh is charged at ₹4; the other $18 - 10 = 8 \text{ kWh}$ is charged at ₹7. Original bill = $10 \times 4 + 8 \times 7 = ₹96$. The cumulative meter total is not the period consumption, and the higher rate does not apply to all 18 kWh. [1 mark]

(iii) Replacement consumption = $18 - 6 = 12 \text{ kWh}$; new bill = $10 \times 4 + 2 \times 7 = ₹54$. The bill falls by $₹96 - ₹54 = ₹42$, not ₹24. Both totals exceed 10 kWh, so all six avoided kWh belong to the ₹7 tier. Equal required output is supplied as evidence, not inferred from power alone. [1 mark]

Mark allocation

(i) Consumed energy is the increase in the cumulative reading: $1268 - 1250 = 18 \text{ kWh}$ (0.5 mark). Average input power during energised hours = $18 \text{ kWh}/30 \text{ h} = 0.60 \text{ kW} = 600 \text{ W}$ (0.5 mark). [1 marks]

(ii) Only the first 10 kWh is charged at ₹4; the other $18 - 10 = 8 \text{ kWh}$ is charged at ₹7. Original bill = $10 \times 4 + 8 \times 7 = ₹96$. The cumulative meter total is not the period consumption, and the higher rate does not apply to all 18 kWh. [1 marks]

(iii) Replacement consumption = $18 - 6 = 12 \text{ kWh}$; new bill = $10 \times 4 + 2 \times 7 = ₹54$. The bill falls by $₹96 - ₹54 = ₹42$, not ₹24. Both totals exceed 10 kWh, so all six avoided kWh belong to the ₹7 tier. Equal required output is supplied as evidence, not inferred from power alone. [1 marks]

Solution 33 — special-39-04-q33

(i) $A = 0.050 \times 10^{-6} = 5.0 \times 10^{-8} \text{ m}^2$. $R_K = R_M = 1.0 \times 10^{-6}/(5.0 \times 10^{-8}) = 20 \Omega$; $R_L = 2.0 \times 10^{-8}/(5.0 \times 10^{-8}) = 0.40 \Omega$. K and M meet the electrical requirement. [1 mark]

(ii) Choose K. L fails the 20Ω requirement at the specified geometry; M meets the resistance requirement but fails the oxidation/durability requirement. [1 mark]

(iii) $A_L = \rho_L L/R = 2.0 \times 10^{-8} \times 1.0/20 = 1.0 \times 10^{-9} \text{ m}^2 = 0.0010 \text{ mm}^2$. This is only 1/50 of the required 0.050 mm^2 , so it violates the fixed-area specification. [1 mark]

Mark allocation

(i) $A = 0.050 \times 10^{-6} = 5.0 \times 10^{-8} \text{ m}^2$. $R_K = R_M = 1.0 \times 10^{-6}/(5.0 \times 10^{-8}) = 20 \Omega$; $R_L = 2.0 \times 10^{-8}/(5.0 \times 10^{-8}) = 0.40 \Omega$. K and M meet the electrical requirement. [1 marks]

(ii) Choose K. L fails the 20Ω requirement at the specified geometry; M meets the resistance requirement but fails the oxidation/durability requirement. [1 marks]

(iii) $A_L = \rho_L L/R = 2.0 \times 10^{-8} \times 1.0/20 = 1.0 \times 10^{-9} \text{ m}^2 = 0.0010 \text{ mm}^2$. This is only 1/50 of the required 0.050 mm^2 , so it violates the fixed-area specification. [1 marks]

Solution 34 — special-39-04-q34

- (i) Original volume = $AL = 0.50 \text{ mm}^2 \times 0.80 \text{ m} = 0.40 \text{ mm}^2 \text{ m} = 4.0 \times 10^{-7} \text{ m}^3$. Conservation gives $A'L' = AL$, or $A' = AL/L'$. [1 mark]
- (ii) Let $k = L'/L$. Then $A' = A/k$, so $R'/R = (L'/L)(A/A') = k^2$ (1 mark). Here $1.00/0.25 = 4$, so $k = 2$ (positive length), $L' = 1.60 \text{ m}$ and $A' = 0.25 \text{ mm}^2$ (1 mark). [2 marks]
- (iii) The design is feasible: $1.60 \text{ m} \leq$ the 1.70 m maximum, and $0.25 \text{ mm}^2 \geq$ the 0.20 mm^2 minimum. Both conditions must hold. [1 mark]
- (iv) At fixed 2.0 V , $P_{\text{original}} = 2.0^2/0.25 = 16 \text{ W}$; $P_{\text{new}} = 2.0^2/1.00 = 4 \text{ W}$. The redesigned wire uses one-quarter the power at the same voltage, not four times as much. [1 mark]

Mark allocation

- (i) Original volume = $AL = 0.50 \text{ mm}^2 \times 0.80 \text{ m} = 0.40 \text{ mm}^2 \text{ m} = 4.0 \times 10^{-7} \text{ m}^3$. Conservation gives $A'L' = AL$, or $A' = AL/L'$. [1 marks]
- (ii) Let $k = L'/L$. Then $A' = A/k$, so $R'/R = (L'/L)(A/A') = k^2$ (1 mark). Here $1.00/0.25 = 4$, so $k = 2$ (positive length), $L' = 1.60 \text{ m}$ and $A' = 0.25 \text{ mm}^2$ (1 mark). [2 marks]
- (iii) The design is feasible: $1.60 \text{ m} \leq$ the 1.70 m maximum, and $0.25 \text{ mm}^2 \geq$ the 0.20 mm^2 minimum. Both conditions must hold. [1 marks]
- (iv) At fixed 2.0 V , $P_{\text{original}} = 2.0^2/0.25 = 16 \text{ W}$; $P_{\text{new}} = 2.0^2/1.00 = 4 \text{ W}$. The redesigned wire uses one-quarter the power at the same voltage, not four times as much. [1 marks]

Solution 35 — special-39-04-q35

- (i) $R_A = 1.2/0.30 = 4.0 \Omega$ and $R_B = 1.2/0.60 = 2.0 \Omega$ (0.5 mark each). [1 mark]
- (ii) $\text{Area}_B/\text{Area}_A = (0.40/0.20)^2 = 4$. Therefore $\rho_B/\rho_A = (R_B/R_A)(\text{Area}_B/\text{Area}_A)(L_A/L_B) = (2/4) \times 4 \times (1/2) = 1$ (1 mark). Equal resistivity means the lower measured resistance of B is explained by its fourfold area and doubled length, not by a lower-resistivity material (1 mark). [2 marks]
- (iii) Unchanged B stock has 2.0Ω per $2.0 \text{ m} = 1.0 \Omega/\text{m}$. A 4.0Ω piece requires 4.0 m , so it cannot meet the 3.0 m limit. [1 mark]
- (iv) At fixed material and length 2.0 m , doubling resistance from 2.0 to 4.0Ω requires half of B's conducting area: $\text{Area}_{\text{new}} = \text{Area}_B/2$. This is a new wire specification, not unchanged stock. [1 mark]

Mark allocation

- (i) $R_A = 1.2/0.30 = 4.0 \Omega$ and $R_B = 1.2/0.60 = 2.0 \Omega$ (0.5 mark each). [1 marks]
- (ii) $\text{Area}_B/\text{Area}_A = (0.40/0.20)^2 = 4$. Therefore $\rho_B/\rho_A = (R_B/R_A)(\text{Area}_B/\text{Area}_A)(L_A/L_B) = (2/4) \times 4 \times (1/2) = 1$ (1 mark). Equal resistivity means the lower measured resistance of B is explained by its fourfold area and doubled length, not by a lower-resistivity material (1 mark). [2 marks]
- (iii) Unchanged B stock has 2.0Ω per $2.0 \text{ m} = 1.0 \Omega/\text{m}$. A 4.0Ω piece requires 4.0 m , so it cannot meet the 3.0 m limit. [1 marks]
- (iv) At fixed material and length 2.0 m , doubling resistance from 2.0 to 4.0Ω requires half of B's conducting area: $\text{Area}_{\text{new}} = \text{Area}_B/2$. This is a new wire specification, not unchanged stock. [1 marks]

Solution 36 — special-39-04-q36

- (i) $\text{Area}_U = \text{volume}/L_U = 4.0 \times 10^{-7} \text{ m}^2$, and $R_U = \rho L_U / \text{Area}_U = 2.5 \, \Omega$ (1 mark). $\text{Area}_V = 2.0 \times 10^{-7} \text{ m}^2$ and $R_V = 1.0 \times 10^{-6} \times 2.0 / (2.0 \times 10^{-7}) = 10 \, \Omega$ (1 mark). The areas differ because the same material volume is distributed over different lengths. [2 marks]
- (ii) $I_U = 10/2.5 = 4.0 \text{ A}$, exceeding 2.0 A ; $I_V = 10/10 = 1.0 \text{ A}$, within the limit. Only V is current-compatible. [1 mark]
- (iii) $P_V = 10 \times 1 = 10 \text{ W}$; $t_V = 6000/10 = 600 \text{ s}$, which is below 700 s . V satisfies both operating requirements. [1 mark]
- (iv) The claim is false in this loss-free model. Either pad transfers the required 6000 J and therefore consumes $6000 \text{ J} = 1/600 \text{ kWh} \approx 0.00167 \text{ kWh}$ for the job. U would be faster (40 W , 150 s) if an adequate source were used, but would not save task energy and is inadmissible on the given source. [1 mark]

Mark allocation

- (i) $\text{Area}_U = \text{volume}/L_U = 4.0 \times 10^{-7} \text{ m}^2$, and $R_U = \rho L_U / \text{Area}_U = 2.5 \, \Omega$ (1 mark). $\text{Area}_V = 2.0 \times 10^{-7} \text{ m}^2$ and $R_V = 1.0 \times 10^{-6} \times 2.0 / (2.0 \times 10^{-7}) = 10 \, \Omega$ (1 mark). The areas differ because the same material volume is distributed over different lengths. [2 marks]
- (ii) $I_U = 10/2.5 = 4.0 \text{ A}$, exceeding 2.0 A ; $I_V = 10/10 = 1.0 \text{ A}$, within the limit. Only V is current-compatible. [1 marks]
- (iii) $P_V = 10 \times 1 = 10 \text{ W}$; $t_V = 6000/10 = 600 \text{ s}$, which is below 700 s . V satisfies both operating requirements. [1 marks]
- (iv) The claim is false in this loss-free model. Either pad transfers the required 6000 J and therefore consumes $6000 \text{ J} = 1/600 \text{ kWh} \approx 0.00167 \text{ kWh}$ for the job. U would be faster (40 W , 150 s) if an adequate source were used, but would not save task energy and is inadmissible on the given source. [1 marks]

Solution 37 — special-39-04-q37

- (i) Subtracting cancels the common contacts: $10.0 - 4.0 = k(1.50 - 0.50)$, so $k = 6.0 \, \Omega/\text{m}$ (1 mark). $C = 4.0 - 6.0 \times 0.50 = 1.0 \, \Omega$ for the two contacts together. Dividing totals by length would give $8.0 \, \Omega/\text{m}$ and $6.67 \, \Omega/\text{m}$ because the fixed contact contribution is incorrectly included as wire resistance (1 mark for C and the misleading-ratio explanation). [2 marks]
- (ii) Predicted total = $6.0 \times 2.00 + 1.0 = 13.0 \, \Omega$. This is a conditional model prediction, not a measurement or proof that all lengths obey it. [1 mark]
- (iii) Repeat 0.50 m while maintaining the contact condition of the $15.0 \, \Omega$ trial. H1 predicts $6.0 \times 0.50 + (1.0 + 2.0) = 6.0 \, \Omega$; H2 predicts the original $4.0 \, \Omega$ because the suspect added segment is absent. Alternatively repeat 1.50 m : H1 predicts $12.0 \, \Omega$, H2 predicts $10.0 \, \Omega$. The single $15.0 \, \Omega$ reading fits both explanations and does not distinguish them (0.5 mark for one repeat with both predictions; 0.5 mark for the non-uniqueness conclusion). [1 mark]

Mark allocation

- (i) Subtracting cancels the common contacts: $10.0 - 4.0 = k(1.50 - 0.50)$, so $k = 6.0 \, \Omega/\text{m}$ (1 mark). $C = 4.0 - 6.0 \times 0.50 = 1.0 \, \Omega$ for the two contacts together. Dividing totals by length would give $8.0 \, \Omega/\text{m}$ and $6.67 \, \Omega/\text{m}$ because the fixed contact contribution is incorrectly included as wire resistance (1 mark for C and the misleading-ratio explanation). [2 marks]
- (ii) Predicted total = $6.0 \times 2.00 + 1.0 = 13.0 \, \Omega$. This is a conditional model prediction, not a measurement or proof that all lengths obey it. [1 marks]

(iii) Repeat 0.50 m while maintaining the contact condition of the 15.0 Ω trial. H1 predicts $6.0 \times 0.50 + (1.0 + 2.0) = 6.0 \Omega$; H2 predicts the original 4.0 Ω because the suspect added segment is absent. Alternatively repeat 1.50 m: H1 predicts 12.0 Ω , H2 predicts 10.0 Ω . The single 15.0 Ω reading fits both explanations and does not distinguish them (0.5 mark for one repeat with both predictions; 0.5 mark for the non-uniqueness conclusion). [1 marks]

Solution 38 — special-39-04-q38

(i) The drawing gives Area_{min} = 0.20 mm² = 2.0×10^{-7} m², path S = 0.80 m and path T = 1.20 m. $R_S = 4.0 \Omega$ and $R_T = 6.0 \Omega$. At 6.0 V, $I_S = 1.5 \text{ A} > 1.2 \text{ A}$, so S is rejected; $I_T = 1.0 \text{ A}$. [1 mark]

(ii) No. Increasing area decreases $R = \rho L/A$ and increases I at fixed voltage. Since S already draws too much current at the smallest permitted area, any permitted larger area makes that failure worse. [1 mark]

(iii) Choose path T with area 0.20 mm². Its current is $6/6 = 1.0 \text{ A} \leq 1.2 \text{ A}$ (1 mark). Its heat in 60 s is $H = VIt = 6 \times 1 \times 60 = 360 \text{ J}$, exactly the minimum, so this design meets both requirements (1 mark). [2 marks]

Mark allocation

(i) The drawing gives Area_{min} = 0.20 mm² = 2.0×10^{-7} m², path S = 0.80 m and path T = 1.20 m. $R_S = 4.0 \Omega$ and $R_T = 6.0 \Omega$. At 6.0 V, $I_S = 1.5 \text{ A} > 1.2 \text{ A}$, so S is rejected; $I_T = 1.0 \text{ A}$. [1 marks]

(ii) No. Increasing area decreases $R = \rho L/A$ and increases I at fixed voltage. Since S already draws too much current at the smallest permitted area, any permitted larger area makes that failure worse. [1 marks]

(iii) Choose path T with area 0.20 mm². Its current is $6/6 = 1.0 \text{ A} \leq 1.2 \text{ A}$ (1 mark). Its heat in 60 s is $H = VIt = 6 \times 1 \times 60 = 360 \text{ J}$, exactly the minimum, so this design meets both requirements (1 mark). [2 marks]

Solution 39 — special-39-04-q39

(i) Use parallel branches across the source, with a separate switch in series with each device within its branch. Each on-device receives 12 V, and opening one branch does not interrupt the other; a shared series path would not meet these requirements. [1 mark]

(ii) $I_{\text{pad}} = 12 \text{ W}/12 \text{ V} = 1.0 \text{ A}$; $I_{\text{indicator}} = 3 \text{ W}/12 \text{ V} = 0.25 \text{ A}$. The source supplies their sum, 1.25 A. [1 mark]

(iii) Original daily energy = $12 \times (20/60) + 3 \times 1 = 7 \text{ Wh}$. New daily energy = $12 \times (10/60) + 3 \times 1 = 5 \text{ Wh}$, so the saving is 2 Wh = 0.002 kWh = 7200 J (1 mark). Percentage saving = $(2/7) \times 100 \approx 28.6\%$, not 50%, because the indicator's energy is unchanged (1 mark). [2 marks]

Mark allocation

(i) Use parallel branches across the source, with a separate switch in series with each device within its branch. Each on-device receives 12 V, and opening one branch does not interrupt the other; a shared series path would not meet these requirements. [1 marks]

(ii) $I_{\text{pad}} = 12 \text{ W}/12 \text{ V} = 1.0 \text{ A}$; $I_{\text{indicator}} = 3 \text{ W}/12 \text{ V} = 0.25 \text{ A}$. The source supplies their sum, 1.25 A. [1 marks]

(iii) Original daily energy = $12 \times (20/60) + 3 \times 1 = 7 \text{ Wh}$. New daily energy = $12 \times (10/60) + 3 \times 1 = 5 \text{ Wh}$, so the saving is 2 Wh = 0.002 kWh = 7200 J (1 mark). Percentage saving = $(2/7) \times 100 \approx$

28.6%, not 50%, because the indicator's energy is unchanged (1 mark). [2 marks]

Source-pattern and rationale appendix

These questions are original. References identify supplied source patterns and curriculum connections, not official provenance for these questions or predictions of future examinations. No missing references or rationale have been inferred.

Question 1 — special-39-04-q01

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.1, printed pp.171–172: current and ammeter connection.

Competency classification: No

Competency marks: 0

Competency rationale: Recall of the definition of current and the standard ammeter connection; no charge-time calculation or competency claim.

Rationale tags: foundational
routine-knowledge

Question 2 — special-39-04-q02

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.2, printed pp.173–174: $V = W/Q$ and $W = VQ$.

Competency classification: No

Competency marks: 0

Competency rationale: A direct two-factor $W = VQ$ ratio comparison, conservatively foundational; not a repeat of computing voltage from a supplied work/charge pair.

Rationale tags: foundational
routine-knowledge

Question 3 — special-39-04-q03

Source patterns: pyq-2023-246eac3017f0793c27bba6a6
ncert-11

NCERT references: NCERT Ch.11 §11.4, Activity 11.1, printed pp.175–176: V/I and graph interpretation.
NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179: R proportional to length, inverse area and material resistivity.

Competency classification: Yes

Competency marks: 1

Competency rationale: Reads the I -vertical graph, then removes the known length difference to reverse the apparent resistance-based material ranking. Equal-geometry comparison is no longer available.

Rationale tags: reasoning
full-competency

Question 4 — special-39-04-q04

Source patterns: cbse_sqp_2018-19_electricity_q13
ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.5, Activity 11.3, printed pp.177–179; Example 11.6, p.180

Competency classification: Yes

Competency marks: 1

Competency rationale: Separates a measurable electrical property from an unsupported identity claim after controlling the geometry ratio.

Rationale tags: reasoning
material-inference

Question 5 — special-39-04-q05

Source patterns: pyq-2024-1e2d386b35acad1e9e07f369
ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.5, Activity 11.3, printed pp.177–179; Example 11.6, p.180

Competency classification: Yes

Competency marks: 1

Competency rationale: Discriminates conducting diameter from an irrelevant outer geometric measurement.

Rationale tags: reasoning
conducting-geometry

Question 6 — special-39-04-q06

Source patterns: ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Class X Science, Ch.11 §11.6.2, Activity 11.6, printed pp.185–187

Competency classification: Yes

Competency marks: 1

Competency rationale: Requires extracting series/parallel connectivity from node dots, then distinguishing source and branch currents.

Rationale tags: reasoning
circuit-topology

Question 7 — special-39-04-q07

Source patterns: ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.1, printed pp.171–172
NCERT Class X Science, Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184
NCERT Class X Science, Ch.11 §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Evaluates a conservation misconception using steady-series-current evidence.

Rationale tags: reasoning
charge-versus-energy

Question 8 — special-39-04-q08

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179: R proportional to length, inverse area and material resistivity.

Competency classification: No

Competency marks: 0

Competency rationale: Direct area-unit conversion; foundational rather than material inference or resistivity-unit recall.

Rationale tags: foundational
routine-knowledge

Question 9 — special-39-04-q09

Source patterns: pyq-2018-ac2b7b28762938a4a2df67a3
ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.4, Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 1

Competency rationale: Compares a repeated observation with a consistent model and limits the conclusion.

Rationale tags: reasoning
experimental-consistency

Question 10 — special-39-04-q10

Source patterns: pyq-2024-1e2d386b35acad1e9e07f369
ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.5, Activity 11.3, printed pp.177–179; Example 11.6, p.180

Competency classification: No

Competency marks: 0

Competency rationale: Direct standard diameter-area-resistance scaling; foundational even though posed backwards.

Rationale tags: foundational
routine-knowledge

Question 11 — special-39-04-q11

Source patterns: ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.7, printed pp.188–189
NCERT Class X Science, Ch.11 §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Identifies the controlled variable that reverses a heating comparison.

Rationale tags: reasoning
heating-constraints

Question 12 — special-39-04-q12

Source patterns: ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.8, printed pp.191–192

Competency classification: No

Competency marks: 0

Competency rationale: Direct commercial-energy unit conversion.

Rationale tags: foundational
energy-units

Question 13 — special-39-04-q13

Source patterns: cbse_sqp_2019-20_electricity_q29
ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.5, Activity 11.3, printed pp.177–179; Example 11.6, p.180
NCERT Class X Science, Ch.11 §11.6.2, Activity 11.6, printed pp.185–187

Competency classification: Yes

Competency marks: 1

Competency rationale: Compares equal and unequal material allocations in parallel and rejects the inference that equal total length fixes source current.

Rationale tags: reasoning
cut-wire-redesign

Question 14 — special-39-04-q14

Source patterns: cbse_sqp_2018-19_electricity_q13
ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.5, Activity 11.3, printed pp.177–179; Example 11.6, p.180

Competency classification: Yes

Competency marks: 1

Competency rationale: Recognises a geometric confounder and chooses information needed to isolate the material property.

Rationale tags: reasoning
fair-material-comparison

Question 15 — special-39-04-q15

Source patterns: ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.4, Activity 11.1, printed pp.175–176
NCERT Class X Science, Ch.11 §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Compares two simultaneous changes and rejects the inference that equal current means equal power.

Rationale tags: reasoning
power-comparison

Question 16 — special-39-04-q16

Source patterns: ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.5, Activity 11.3, printed pp.177–179; Example 11.6, p.180

Competency classification: Yes

Competency marks: 1

Competency rationale: Distinguishes occupied space and visual shape from actual conducting-path geometry.

Rationale tags: reasoning
shape-and-resistance

Question 17 — special-39-04-q17

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179: R proportional to length, inverse area and material resistivity.

Competency classification: Yes

Competency marks: 1

Competency rationale: Uses explicit candidate evidence to distinguish exclusion from unique identification; no stretch-factor calculation or rehearsal of the conserved-volume long task.

Rationale tags: reasoning
full-competency

Question 18 — special-39-04-q18

Source patterns: ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.1, printed pp.171–172
NCERT Class X Science, Ch.11 §11.2, printed p.173

Competency classification: No

Competency marks: 0

Competency rationale: Checks two familiar facts and their explanatory relation; conservatively foundational.

Rationale tags: foundational
measurement-and-units

Question 19 — special-39-04-q19

Source patterns: pyq-2024-1e2d386b35acad1e9e07f369
ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.5, Activity 11.3, printed pp.177–179; Example 11.6, p.180

Competency classification: No

Competency marks: 0

Competency rationale: Familiar squared-diameter misconception; conservatively foundational like Q10.

Rationale tags: foundational
routine-knowledge

Question 20 — special-39-04-q20

Source patterns: ncert-11

NCERT references: NCERT Class X Science, Ch.11 §11.6.2, Activity 11.6, printed pp.185–187

Competency classification: Yes

Competency marks: 1

Competency rationale: Separates total-current change from branch-voltage behaviour under a fixed-source constraint.

Rationale tags: reasoning
parallel-branch-change

Question 21 — special-39-04-q21

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.1, printed pp.171–172

Competency classification: Yes

Competency marks: 1

Competency rationale: Only part (ii) evaluates an invalid comparison; part (i) is routine substitution.

Rationale tags: reasoning
partial-competency

Question 22 — special-39-04-q22

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.2, printed p.173

Competency classification: Yes

Competency marks: 1

Competency rationale: Only (ii) transfers the result to an equal-charge comparison; (i) is direct division and ratio.

Rationale tags: reasoning
partial-competency

Question 23 — special-39-04-q23

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.2, printed p.173

NCERT Ch.11 §11.4, Activity 11.1, printed pp.175–176

NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184

Competency classification: Yes

Competency marks: 2

Competency rationale: Both marks require diagnosing the actual instrument span, not merely substituting nominal meter readings.

Rationale tags: reasoning
full-competency

Question 24 — special-39-04-q24

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179; Examples 11.5–11.6, p.180

Competency classification: No

Competency marks: 0

Competency rationale: Direct formula, meaning and unit recall; no competency marks assigned.

Rationale tags: foundational
routine-knowledge

Question 25 — special-39-04-q25

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.7, printed pp.188–189

Competency classification: Yes

Competency marks: 1

Competency rationale: Part (ii) reasons across two changed variables; part (i) is foundational calculation.

Rationale tags: reasoning
partial-competency

Question 26 — special-39-04-q26

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.4, printed p.176: constant-temperature condition; §11.5, printed p.179: resistance and resistivity vary with temperature.

Competency classification: Yes

Competency marks: 1

Competency rationale: Only (ii) diagnoses temperature confounding and proposes a repeat; (i) is routine ratio calculation.

Rationale tags: reasoning
partial-competency

Question 27 — special-39-04-q27

Source patterns: cbse_sqp_2018-19_electricity_q13
ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179; Examples 11.5–11.6, p.180

Competency classification: Yes

Competency marks: 2

Competency rationale: Count (i) geometry-normalised pair selection and (iii) diagnosis; (ii) is a routine ratio from computed values.

Rationale tags: reasoning
partial-competency

Question 28 — special-39-04-q28

Source patterns: pyq-2018-ac2b7b28762938a4a2df67a3
SQP_2025-26_q36
ncert-11

NCERT references: NCERT Ch.11 §11.4, Activity 11.1, printed pp.175–176

NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179; Examples 11.5–11.6, p.180

Competency classification: Yes

Competency marks: 2

Competency rationale: Count (i) model consistency and the one-mark admissibility decision in (ii); candidate-current calculations are foundational.

Rationale tags: reasoning
partial-competency

Question 29 — special-39-04-q29

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.4, Activity 11.1, printed pp.175–176

NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184

NCERT Ch.11 §11.7, printed pp.188–189

Competency classification: Yes

Competency marks: 1

Competency rationale: Only (iii) challenges a heat misconception; routine series-voltage subtraction and Ohm-law calculation in (i)–(ii) are foundational.

Rationale tags: reasoning
partial-competency

Question 30 — special-39-04-q30

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179: R proportional to length, inverse area and material resistivity.

NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184: common series current and additive resistance.

NCERT Ch.11 §11.7, printed pp.188–189: $H = I^2 R t$.

NCERT Ch.11 §11.8, printed pp.191–192: $P = I^2 R$, energy, kWh and cost.

Competency classification: Yes

Competency marks: 2

Competency rationale: Part (i) is routine resistance/heating arithmetic. Count the spatial heat-localisation decision and common-current explanation in (ii), plus the limit of the temperature inference in (iii).

Rationale tags: reasoning
partial-competency

Question 31 — special-39-04-q31

Source patterns: pyq-2023-246eac3017f0793c27bba6a6

SQP_2025-26_q36

ncert-11

NCERT references: NCERT Ch.11 §11.4, Activity 11.1, printed pp.175–176

NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179; Examples 11.5–11.6, p.180

Competency classification: Yes

Competency marks: 2

Competency rationale: Part (i) is a straightforward graph read-off, conservatively foundational. Parts (ii)–(iii) normalise geometry and apply a length constraint to choose stock.

Rationale tags: reasoning
partial-competency

Question 32 — special-39-04-q32

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.8, printed pp.191–192: $P = I^2 R$, energy, kWh and cost.

Competency classification: Yes

Competency marks: 1

Competency rationale: Parts (i) and (ii) are foundational meter/power and explicitly specified tariff arithmetic. Count only (iii), which evaluates the manager's wrong-tier claim against the changed total-consumption baseline.

Rationale tags: reasoning
partial-competency

Question 33 — special-39-04-q33

Source patterns: pyq-2022-cfec2ef8067dd02793a5ebf2
SQP_2025-26_q36
ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179; Examples 11.5–11.6, p.180
NCERT Ch.11 §11.5, printed p.179; §11.7.1, p.190 (heating elements)

Competency classification: Yes

Competency marks: 2

Competency rationale: Count (ii) joint electrical/durability screening and (iii) inadmissible-workaround assessment; (i) is routine candidate substitution.

Rationale tags: reasoning
partial-competency

Question 34 — special-39-04-q34

Source patterns: pyq-2025-89ec80e87c4adf229cc3e824
SQP_2025-26_q36
ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179; Examples 11.5–11.6, p.180
NCERT Ch.11 §11.7, printed pp.188–189
NCERT Ch.11 §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 3

Competency rationale: Count the two-mark conserved-volume inverse design in (ii) and feasibility in (iii); direct volume and power calculations in (i)/(iv) are foundational.

Rationale tags: reasoning
partial-competency

Question 35 — special-39-04-q35

Source patterns: cbse_sqp_2018-19_electricity_q13
pyq-2024-1e2d386b35acad1e9e07f369
SQP_2025-26_q36
ncert-11

NCERT references: NCERT Ch.11 §11.4, Activity 11.1, printed pp.175–176
NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179; Examples 11.5–11.6, p.180

Competency classification: Yes

Competency marks: 3

Competency rationale: Count geometry-normalised inference in (ii) and length-limit decision in (iii); direct resistance and fixed-length inverse-area scaling in (i)/(iv) are foundational.

Rationale tags: reasoning
partial-competency

Question 36 — special-39-04-q36

Source patterns: pyq-2025-89ec80e87c4adf229cc3e824
sqp_2021-22_term2_q11
ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179; Examples 11.5–11.6, p.180
NCERT Ch.11 §11.7, printed pp.188–189
NCERT Ch.11 §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 3

Competency rationale: Parts (i) use routine $A = \text{volume}/L$ then $R = \rho L/A$ and receive no competency credit. Count current-limit decision (ii), deadline check (iii), and energy-claim evaluation (iv).

Rationale tags: reasoning
partial-competency

Question 37 — special-39-04-q37

Source patterns: ncert-11

NCERT references: NCERT Ch.11 §11.4, Activity 11.1, printed pp.175–176: V/I and graph interpretation.
NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179: R proportional to length, inverse area and material resistivity.
NCERT Ch.11 §11.6.1, Activities 11.4–11.5, printed pp.182–184: common series current and additive resistance.

Competency classification: Yes

Competency marks: 3

Competency rationale: Count two marks in (i) for difference-based cancellation of an unknown common series contribution and diagnosis of the length-ratio trap, plus (iii) for a discriminating repeat with competing predictions. The direct model prediction in (ii) is foundational.

Rationale tags: reasoning
partial-competency

Question 38 — special-39-04-q38

Source patterns: SQP_2025-26_q36
sqp_2021-22_term2_q11
ncert-11

NCERT references: NCERT Ch.11 §11.5, Activity 11.3, printed pp.177–179; Examples 11.5–11.6, p.180
NCERT Ch.11 §11.7, printed pp.188–189
NCERT Ch.11 §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 3

Competency rationale: Count the rejection decision (i), monotonic rescue test (ii), and final heat/feasibility check (one mark in iii); repeating the already calculated current in (iii) earns no competency credit.

Rationale tags: reasoning
partial-competency

Question 39 — special-39-04-q39

Source patterns: sqp_2021-22_term2_q11
ncert-11

NCERT references: NCERT Ch.11 §11.6.2, Activity 11.6, printed pp.185–187
NCERT Ch.11 §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 2

Competency rationale: Count topology choice with functional explanation (i) and total-baseline percentage interpretation (one mark in iii); current and saved-energy arithmetic are foundational.

Rationale tags: reasoning
partial-competency

Source reference details

id: SQP_2025-26_q36

title: CBSE Class X Science (086) Sample Question Paper 2025-26 (English)

url: https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 36

session: 2025-26

exam_type: SQP

original_section: Section – C

language: en

pages: 13

page: 13

bank_verified: Yes

bank_answer_verified: Yes

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: cbse_sqp_2018-19_electricity_q13

title: Class-X Science-086 SAMPLE QUESTION PAPER-19

url: https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_SQP_2018-19.pdf

source_role: design-pattern

local_source_record: research/sqp/question_bank.json

question_number: 13

session: 2018-19

exam_type: SQP

original_section: SECTION C

language: en

pages: 3

page: 3

bank_verified: Yes

bank_answer_verified: Yes

bank_answer_source_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: cbse_sqp_2019-20_electricity_q29

title: Class-X Science-086 SAMPLE QUESTION PAPER 2019-20

url: https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_SQP.pdf
source_role: design-pattern
local_source_record: research/sqp/question_bank.json
question_number: 29
session: 2019-20
exam_type: SQP
original_section: SECTION C
language: en
pages: 9
page: 9
bank_verified: Yes
bank_answer_verified: Yes
bank_answer_source_verified: Yes
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: ncert-11
title: NCERT Science, Class X, Chapter 11: Electricity
url: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>
source_role: conceptual-scope
local_source_record: research/sqp/archive/jesc111.txt
notes: Conceptual scope and activity reference only; not provenance of original questions. No question-bank verification flags are inferred.
id: pyq-2018-ac2b7b28762938a4a2df67a3
title: CBSE — Class X Science — 2018 — 31/1 — main
url: https://www.cbse.gov.in/curric~1/qpms2018/qp_10/Science.zip
source_role: design-pattern
local_source_record: research/pyq/question_bank.json
question_number: 27
year: 2018
session: main
exam_type: PYQ
paper_code: 31/1
language: English
source_title_generated: Yes
session_inferred: Yes
original_section: Section - B
pages: 12
page: 12
bank_verified: Yes
bank_answer_verified: No
bank_official_source_verified: Yes
bank_text_verified: Yes
bank_visual_verified: Yes
bank_year_verified: Yes
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: pyq-2022-cfec2ef8067dd02793a5ebf2
title: CBSE — Class X Science — 2022 — main — 31/1/2
url: <https://www.cbse.gov.in/cbsenew/question-paper/2022/X/Scince.zip>
source_role: design-pattern
local_source_record: research/pyq/question_bank.json
question_number: 10
year: 2022
session: main
exam_type: PYQ

paper_code: 31/1/2
term: II
language: English
source_title_generated: Yes
original_section: SECTION B
pages: 7
page: 7
bank_verified: Yes
bank_answer_verified: No
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: pyq-2023-246eac3017f0793c27bba6a6
title: CBSE — Class X Science — 2023 — main — 31/2/1
url: <https://www.cbse.gov.in/cbsenew/question-paper/2023/X/SCIENCE.zip>
source_role: design-pattern
local_source_record: research/pyq/question_bank.json
question_number: 25
year: 2023
session: main
exam_type: PYQ
paper_code: 31/2/1
language: English
source_title_generated: Yes
original_section: SECTION B
pages: 15
page: 15
bank_verified: Yes
bank_answer_verified: No
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: pyq-2024-1e2d386b35acad1e9e07f369
title: CBSE — Class X Science — 2024 — main — 31/4/1
url: <https://www.cbse.gov.in/cbsenew/question-paper/2024/X/SCIENCE.zip>
source_role: design-pattern
local_source_record: research/pyq/question_bank.json
question_number: 26
year: 2024
session: main
exam_type: PYQ
paper_code: 31/4/1
language: English
source_title_generated: Yes
original_section: SECTION B
pages: 13
page: 13
bank_verified: Yes
bank_answer_verified: No
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: pyq-2025-89ec80e87c4adf229cc3e824
title: CBSE — Class X Science — 2025 — main — 31/3/1
url: https://www.cbse.gov.in/cbsenew/question-paper/2025/X/086_Science.zip
source_role: design-pattern
local_source_record: research/pyq/question_bank.json
question_number: 13

year: 2025
session: main
exam_type: PYQ
paper_code: 31/3/1
language: English
source_title_generated: Yes
original_section: SECTION A
pages: 9
page: 9
bank_verified: Yes
bank_answer_verified: No
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.
id: sqp_2021-22_term2_q11
title: CBSE Class X Science (086) Sample Question Paper 2021-22 — Term 2
url: https://cbseacademic.nic.in/web_material/SQP/ClassX_2021_22/Science-SQP_Term2.pdf
source_role: design-pattern
local_source_record: research/sqp/question_bank.json
question_number: 11
session: 2021-22
exam_type: SQP
original_section: Section B
term: 2
language: en
pages: 4
page: 4
bank_verified: Yes
bank_answer_verified: Yes
bank_answer_source_verified: Yes
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.